Correlation between Fear of Movement (Kinesiophobia) and Stair-Climbing Ability following Unilateral Total Knee Replacement: A Cross-Sectional Study

A Dissertation

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ABSTRACT

Background: Total Knee Replacement (TKR) reliably relieves pain and corrects deformity in end-stage knee osteoarthritis, yet many patients continue to struggle with demanding tasks such as stair negotiation. Fear of movement (kinesiophobia), explained by the fear-avoidance model, may contribute to this limitation independently of physical impairment. The relationship between kinesiophobia and objectively measured stair-climbing ability after unilateral TKR has received little attention, particularly in the Indian population.

Objective: To determine the correlation between kinesiophobia and stair-climbing ability in individuals following unilateral TKR, and to describe the levels of kinesiophobia and stair-climbing performance in this population.

Methods: A cross-sectional study was conducted on [n = 50] individuals aged 50–75 years who were 4 weeks to 6 months after unilateral TKR, recruited by convenience sampling from [study setting]. Kinesiophobia was assessed using the 17-item Tampa Scale for Kinesiophobia (TSK-17) and stair-climbing ability using the timed Stair Climb Test (SCT). Normality was tested using the Shapiro–Wilk test, and the association between TSK-17 and SCT scores was analysed using [Pearson's / Spearman's] correlation coefficient. The level of significance was set at $p < 0.05$.

Results: The mean age of participants was [__ ± __] years and the mean TSK-17 score was [__ ± __]; [__ (__%)] participants had high kinesiophobia (TSK-17 > 37). The mean SCT time was [__ ± __] seconds. A [strength and direction] correlation was found between TSK-17 and SCT time (r = [__], p = [__]).

Conclusion: [Based on the results: e.g., "Higher fear of movement was significantly associated with poorer stair-climbing ability after unilateral TKR, indicating that kinesiophobia should be screened for and addressed in post-operative rehabilitation."]

Keywords: Kinesiophobia; Fear of movement; Fear-avoidance model; Total knee replacement; Total knee arthroplasty; Stair climbing; Tampa Scale for Kinesiophobia; Stair Climb Test

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LIST OF ABBREVIATIONS

Abbreviation	Full form
ACL	Anterior Cruciate Ligament
ADL	Activities of Daily Living
AMI	Arthrogenic Muscle Inhibition
BMI	Body Mass Index
CBT	Cognitive Behavioural Therapy
EMG	Electromyography
FAM	Fear-Avoidance Model
ICC	Intraclass Correlation Coefficient
KOOS	Knee Injury and Osteoarthritis Outcome Score
LEFS	Lower Extremity Functional Scale
OA	Osteoarthritis
OARSI	Osteoarthritis Research Society International
OSA	Occupational Self-Assessment
PROM	Patient-Reported Outcome Measure
PSEQ	Pain Self-Efficacy Questionnaire
ROM	Range of Motion
SCT	Stair Climb Test
SD	Standard Deviation
SPSS	Statistical Package for the Social Sciences
STV	Stance Time Variability
TKA	Total Knee Arthroplasty
TKR	Total Knee Replacement
TSK	Tampa Scale for Kinesiophobia
TUG	Timed Up and Go test
WOMAC	Western Ontario and McMaster Universities Osteoarthritis Index
2-MWT / 6MWT	Two-Minute / Six-Minute Walk Test

CHAPTER 1: INTRODUCTION

1.1 Background

Knee osteoarthritis (OA) is one of the most common, disabling and progressive musculoskeletal disorders affecting the ageing population worldwide. It is characterised by progressive degeneration of articular cartilage, subchondral bone sclerosis, osteophyte formation and low-grade synovial inflammation, which together produce pain, stiffness, deformity and a marked decline in quality of life. When conservative measures such as analgesic and anti-inflammatory medication, intra-articular injections, physiotherapy, weight management and lifestyle modification no longer provide adequate relief of pain or improvement in function, Total Knee Arthroplasty (TKA), also termed Total Knee Replacement (TKR), is regarded as the gold-standard surgical treatment (Steinhaus et al., 2017).

Large numbers of TKRs are performed every year with the aims of relieving arthritic pain, correcting frontal-plane deformity and restoring the mechanical stability of the joint, and the demand for the procedure is projected to rise steadily as populations age (Kurtz et al., 2007). From a biomechanical and engineering standpoint, contemporary implants and surgical techniques achieve very high rates of success: most patients have satisfactory radiological alignment, a stable prosthetic joint and a substantial reduction in pain at rest after surgery (Goldberg, 2007).

Despite these structural achievements, there is a well-recognised gap between surgical success and patient-perceived outcome. Approximately one in five patients report dissatisfaction after primary TKR (Bourne et al., 2010), and a considerable proportion continue to experience functional limitations during the first post-operative year even after an uncomplicated procedure. Many of these individuals demonstrate lower physical performance than age-matched peers without knee pathology. Level walking usually recovers to a satisfactory degree early in rehabilitation, whereas more complex locomotor tasks, which demand greater neuromotor coordination, eccentric muscle control and multi-joint movement, often remain impaired for a prolonged period (Dietz et al., 1995).

1.2 Stair Climbing after Total Knee Replacement

Ascending and descending stairs are among the most challenging, and psychologically taxing, activities that a patient must regain after TKR. The ability to negotiate stairs independently is far more than a physical milestone: it is a marker of independence that allows a person to move in and out of the home, access public spaces and transport, and live comfortably in multi-level dwellings (Karunaratne et al., 2024).

Biomechanical studies show that stair ambulation requires substantially greater lower-limb joint excursion, peak joint loading and muscular effort than level walking. During stair ascent the quadriceps and gluteal muscles must generate large concentric forces to raise the body weight against gravity from knee flexion angles that commonly exceed 90–100 degrees (Chen et al., 2021). Stair descent presents a different challenge, demanding finely graded eccentric contraction of the quadriceps to brake and control the downward acceleration of the body's centre of mass. Even minor deficits in neuromuscular coordination or proprioception therefore become evident during descent, which places additional stress on the prosthetic knee and provokes discomfort. Because of these high mechanical demands, stair negotiation has been described as the activity that causes the greatest difficulty for patients after unilateral TKR and is closely linked to fear and dissatisfaction (Ma et al., 2025).

Healthy individuals negotiate stairs with a reciprocal (step-over-step) pattern, placing alternate feet on successive steps. Following unilateral TKR, many patients adopt a non-reciprocal step-to (step-by-step) pattern in which the same limb always leads and the other follows. The non-operated limb typically leads during ascent and the operated limb during descent, so that the replaced knee is spared the most demanding concentric and eccentric loading. Although this strategy protects the operated knee, it imposes asymmetric loads on the lumbo-pelvic-hip complex and the contralateral limb. Patients also rely heavily on handrails, which unload the lower limbs, improve balance and reduce the risk of falls. While handrail use provides essential stability early after surgery, prolonged dependence on it is indicative of incomplete rehabilitation (Zeni & Snyder-Mackler, 2010a).

1.3 Biomedical Determinants of Stair-Climbing Ability

Traditionally, impaired stair negotiation after joint replacement has been studied from a biomedical perspective. The literature has concentrated on physical determinants such as active knee range of motion, prosthetic alignment, implant design and, above all, quadriceps weakness (Christensen et al., 2018). Arthrogenic muscle inhibition (AMI), an ongoing reflex inhibition of the muscles surrounding a joint after trauma or surgery, contributes to profound and long-lasting quadriceps atrophy and weakness after TKR.

Structural impairment, stiffness and quadriceps weakness clearly play important roles in recovery. However, multivariate analyses have repeatedly shown that physical deficits explain only part of the variance in stair-climbing ability after TKR. Patients with similar quadriceps strength, prosthesis design and knee flexion range display strikingly different stair performance: some climb efficiently with a reciprocal pattern, whereas others move hesitantly and rigidly and depend entirely on the handrail. These

observations expose the limitations of a purely biomedical approach and point to the role of other mediating factors (Vinueza-Veloz et al., 2014).

1.4 The Fear-Avoidance Model

Contemporary rehabilitation science increasingly views recovery from major reconstructive joint surgery as a psycho-physical process and recognises the value of incorporating cognitive-behavioural concepts. The fear-avoidance model (FAM) has emerged as a leading framework for explaining how cognitive and emotional factors influence the transition from acute pain to persistent pain and disability (Lethem et al., 1983; Vlaeyen & Linton, 2000; Wideman et al., 2013).

Originally developed to explain why some individuals with low back pain become chronically disabled, the model proposes that the way a person interprets pain determines the path of recovery. Acute pain, such as the intense inflammatory pain that follows TKR, can lead the patient along one of two pathways. When the patient appraises the pain as a normal, predictable and temporary consequence of tissue healing, fear remains low and the patient *confronts* the situation by participating actively in progressive physiotherapy, adhering to home exercise and gradually resuming more demanding activities. This adaptive pathway supports optimal physical and structural recovery (Kakavas et al., 2020).

When the appraisal is negative, a self-reinforcing maladaptive cycle begins. Its starting point is **pain catastrophising**, an exaggerated negative cognitive and emotional orientation towards actual or anticipated pain. Patients who catastrophise interpret their post-operative pain as highly threatening, uncontrollable and a sign of serious structural failure, such as wound breakdown or prosthetic loosening. This cognitive magnification amplifies central processing of nociceptive input and lowers pain thresholds (Tidmarsh et al., 2022).

Pain catastrophising is a direct antecedent of **kinesiophobia**, defined by Kori, Miller and Todd (1990) as “an excessive, irrational and debilitating fear of physical movement and activity resulting from a feeling of vulnerability to painful injury or re-injury”. The patient holds the belief that exercise, exertion or movement will cause re-injury, irreversible damage or more pain (Zale & Ditre, 2015). Within the FAM, kinesiophobia is expressed through avoidance and hypervigilance: patients avoid movements expected to cause discomfort, limit their engagement with exercise and withdraw from valued activities. The consequences are disuse, depression and disability. Lack of loading leads to rapid muscle atrophy, joint stiffness, soft-tissue contracture and reduced cardiovascular fitness (Bortz, 1984). This physical deconditioning, combined with the distress of lost independence, paradoxically increases disability and pain,

appears to confirm the catastrophic beliefs, and locks the patient in a vicious cycle of fear and decline (Leeuw et al., 2007). The model as applied to TKR is summarised in Figure 1.1.

Figure 1.1: The fear-avoidance model applied to recovery after TKR

Adaptive (confrontation) pathway	Maladaptive (avoidance) pathway
Post-operative pain after TKR	Post-operative pain after TKR
↓ Pain appraised as normal and temporary	↓ Pain catastrophising (pain seen as a sign of damage)
↓ Low fear of movement	↓ Kinesiophobia (fear of movement / re-injury)
↓ Confrontation: active participation in exercise and graded return to stairs	↓ Avoidance and hypervigilance: step-to pattern, handrail dependence, offloading of operated knee
↓ Recovery of strength, ROM and motor control	↓ Disuse, deconditioning, depression and disability
↓ Independent, reciprocal stair negotiation	↓ Poor stair-climbing ability, which reinforces pain and fear

1.5 Kinesiophobia after Total Knee Replacement

Kinesiophobia is no longer regarded as confined to chronic low back pain or fibromyalgia. It is highly prevalent in acute orthopaedic populations undergoing major joint reconstruction such as TKR. Pooled estimates suggest a prevalence of approximately 35% after TKR, with some studies reporting rates as high as 56.67% (Desai et al., 2016; Singh et al., 2025).

This high prevalence reflects a distinctive feature of TKR rehabilitation: the operated joint must be actively mobilised and loaded almost immediately to prevent arthrofibrosis and adhesions. For a patient who has not been psychologically prepared before surgery, bending and bearing weight on a freshly operated, swollen and painful knee is sufficient to provoke an intense fear response (Shang et al., 2026).

The effects of kinesiophobia are particularly evident during complex and demanding tasks such as stair negotiation. When a fearful patient approaches a staircase, the limiting factor may not be weak muscles but fear-driven motor control. During stair descent, fear tends to cause excessive activation of antagonists such as the hamstrings and gastrocnemius, producing co-contraction around the knee to brace it against an

anticipated collapse, fall or pain. This muscular stiffening prevents the normal kinematics of descent, as the patient cannot allow the knee to flex freely, and results in a slow, hesitant swing phase. The patient relies on the handrail not because strength is insufficient but because of fear, and persistent avoidance prevents the re-learning of eccentric motor control (Tiedemann et al., 2007).

1.6 Unilateral versus Bilateral TKR

The expression of kinesiophobia appears to be influenced by whether the patient has undergone unilateral or bilateral TKR. Comparative cross-sectional studies have found higher rates of kinesiophobia after unilateral than after bilateral TKR (34.1% versus 22.5%) (Kazarian et al., 2021).

This difference has an important biomechanical and behavioural explanation. A patient with unilateral TKR has a non-operated limb that is relatively strong and pain-free, and therefore becomes the preferred limb for demanding tasks; mechanical demands can simply be shifted onto it (Song et al., 2024). Such offloading is a powerful avoidance strategy: the patient never has to learn to load the operated knee eccentrically during stair descent because the non-operated limb can always be used instead. This behaviour reinforces the fear-avoidance cycle and allows kinesiophobia to become entrenched. Patients with bilateral TKR have no pain-free limb on which to rely and are compelled to load both knees. For this reason, the present study focuses specifically on patients with unilateral TKR.

1.7 Rationale for the Present Study

Although the role of psychological factors in self-reported outcomes after TKR is increasingly acknowledged, the relationship between kinesiophobia and objectively measured stair performance remains poorly defined in routine clinical practice. Few studies have linked the Tampa Scale for Kinesiophobia (TSK) with performance-based measures of stair negotiation (Kortlever et al., 2020).

Understanding this relationship is clinically important because conventional physiotherapy programmes that address only biomechanical impairments may overlook the cognitive component of disability. If a patient’s difficulty on stairs is driven more by persistent fear of movement than by actual muscle weakness, quadriceps strengthening and joint mobilisation alone are unlikely to be sufficient; instead, a holistic approach that integrates cognitive strategies with physical rehabilitation is required (De Vroey et al., 2020). The present study therefore sets out to quantify the association between kinesiophobia and stair-climbing ability in individuals in the early phase of rehabilitation after unilateral TKR. By establishing whether psychological

characteristics are associated with functional mobility independently of the joint, this work aims to encourage clinicians to look beyond the knee to the person who moves it (Neelaraju et al., 2023).

1.8 Need of the Study

TKR is one of the most frequently performed procedures for advanced knee OA and other degenerative joint diseases. Although it is highly effective in relieving pain and restoring alignment, many patients continue to experience difficulty with functional tasks such as climbing stairs, rising from a chair and walking on uneven ground. Of these, stair climbing is among the most difficult and the most important for independent living.

Rehabilitation after unilateral TKR conventionally centres on physical impairments: pain relief, restoration of range of motion, lower-limb strengthening and gait training. Recovery, however, does not depend on physical factors alone. Psychological factors, and kinesiophobia in particular, may strongly influence a patient's engagement in rehabilitation and in functional activities. Patients with high kinesiophobia are reluctant to undertake weight-bearing tasks and restrict their movement for fear of pain or injury.

Despite growing recognition of psychological influences in musculoskeletal rehabilitation, little attention has been given to the association between kinesiophobia and stair climbing after unilateral TKR. Existing research has concentrated on overall function and pain rather than on specific, high-demand tasks that are vital for independence and social participation. Moreover, very little research has been conducted in the Indian population, in whom cultural factors, lifestyle demands (such as floor-level activities and multi-storey housing without lifts), access to rehabilitation and beliefs about pain and movement may differ and may therefore influence recovery differently.

The findings of this study can help physiotherapists to identify patients with high kinesiophobia and to design rehabilitation programmes that address both the psychological and physical barriers to recovery. Early identification and management of kinesiophobia may lead to better stair-climbing performance and greater functional independence.

1.9 Significance of the Study

1. **Clinical significance:** The study will inform physiotherapists and other rehabilitation professionals about the influence of fear of movement on stair-climbing ability after unilateral TKR. Understanding psychological barriers will

enable clinicians to develop more comprehensive, biopsychosocial rehabilitation programmes.

2. **Functional significance:** Stair climbing is an essential activity of daily living and a key determinant of functional independence after TKR. The findings will clarify how kinesiophobia relates to functional mobility after surgery.
3. **Research significance:** Evidence on the relationship between kinesiophobia and stair-climbing ability after unilateral TKR is limited, particularly in India. This study will add to the existing literature and provide a basis for future longitudinal and interventional research.

1.10 Research Gap

1. **Limited studies specifically correlating kinesiophobia with stair-climbing ability after unilateral TKR.** Most post-TKR rehabilitation research focuses on pain management, range of motion, muscle strength or general functional mobility; little has examined fear of movement specifically in relation to stair negotiation in unilateral TKR.
2. **Emphasis on general functional outcomes rather than task-specific activities.** Much of the literature examines walking speed, balance or composite outcome scores such as WOMAC or KOOS, while stair climbing has been relatively neglected.
3. **Few studies combine objective stair-climbing measurement with a validated kinesiophobia scale.** Earlier studies have mainly relied on questionnaires; few have combined a valid fear measure such as the TSK with an objective performance test such as the Stair Climb Test (SCT).
4. **Limited evidence from the Indian population.** Most available studies come from Western populations. Differences in culture, pain perception, rehabilitation facilities and living conditions may influence outcomes in Indian patients.

1.11 Statement of the Problem

Despite successful surgery, many individuals who have undergone unilateral TKR continue to have difficulty negotiating stairs. Whether, and to what extent, fear of movement is associated with this difficulty has not been adequately established. The present study addresses the following research questions:

1. Is there a relationship between fear of movement (kinesiophobia) and stair-climbing ability among patients with unilateral TKR?

2. Do patients with high fear of movement have greater difficulty in stair climbing after unilateral TKR?
3. What is the level of fear of movement among patients with unilateral TKR?
4. How does stair-climbing ability differ according to the level of fear of movement among patients with unilateral TKR?

1.12 Aim and Objectives

Aim

To study the relationship between fear of movement (kinesiophobia) and stair-climbing ability in individuals following unilateral Total Knee Replacement.

Primary objective

1. To determine the correlation between fear of movement (kinesiophobia) and stair-climbing ability following unilateral TKR.

Secondary objectives

1. To assess the level of kinesiophobia in individuals following unilateral TKR.
2. To evaluate stair-climbing ability in individuals following unilateral TKR.
3. To analyse whether higher levels of kinesiophobia are associated with poorer stair-climbing performance after unilateral TKR.

1.13 Hypotheses

Null hypothesis (H_0): There is no significant correlation between fear of movement (kinesiophobia) and stair-climbing ability following unilateral TKR.

Alternative hypothesis (H_1): There is a significant correlation between fear of movement (kinesiophobia) and stair-climbing ability following unilateral TKR.

1.14 Operational Definitions

Total Knee Replacement (TKR): A surgical procedure in which the articular surfaces of the femur, tibia and (optionally) patella are replaced with prosthetic components. *Unilateral TKR* refers to replacement of one knee only.

Kinesiophobia: An excessive, irrational and debilitating fear of physical movement and activity resulting from a feeling of vulnerability to painful injury or re-injury (Kori et al., 1990). In this study it is quantified by the total score on the 17-item Tampa Scale for Kinesiophobia (TSK-17); a score above 37 is considered high kinesiophobia.

Stair-climbing ability: The capacity to ascend and descend a flight of stairs safely and efficiently. In this study it is quantified as the time in seconds taken to complete the Stair Climb Test (SCT); a longer time indicates poorer ability.

Reciprocal stair pattern: Ascending or descending with alternate feet placed on successive steps (step-over-step).

Step-to pattern: Ascending or descending by placing both feet on each step before proceeding to the next (non-reciprocal).

Post-operative rehabilitation phase: The period from 4 weeks to 6 months after surgery during which the participant is receiving or has recently received physiotherapy.

1.15 Scope and Delimitations

- The study is delimited to individuals aged 50–75 years who have undergone primary unilateral TKR and are 4 weeks to 6 months post-operative.
- Kinesiophobia is assessed with the TSK-17 and stair-climbing ability with the timed SCT; other psychological constructs (anxiety, depression, catastrophising, self-efficacy) are not formally measured.
- Participants are recruited from [name of hospital(s) / rehabilitation centre(s)] in Jaipur, Rajasthan, which may limit generalisability.

CHAPTER 2: REVIEW OF LITERATURE

This chapter reviews the literature relevant to the present study under the following themes: (i) the fear-avoidance model and the concept of kinesiophobia; (ii) stair negotiation and its biomechanics after TKR; (iii) prevalence and determinants of kinesiophobia after TKR; (iv) the influence of kinesiophobia on functional outcomes after TKR; and (v) the outcome measures used in this study.

2.1 The Fear-Avoidance Model and Kinesiophobia

Lethem et al. (1983) first proposed the "fear-avoidance" model of exaggerated pain perception, suggesting that individuals respond to pain on a continuum between confrontation and avoidance, and that avoidance is associated with the persistence of pain behaviour and disability.

Kori, Miller and Todd (1990) introduced the term *kinesiophobia* to describe an excessive, irrational and debilitating fear of movement and activity arising from a feeling of vulnerability to painful injury or re-injury, and developed the Tampa Scale for Kinesiophobia (TSK) to measure it.

Vlaeyen et al. (1995) examined fear of movement/(re)injury in chronic low back pain using a Dutch version of the TSK and demonstrated that pain-related fear was associated with poorer behavioural performance, independently of pain intensity. They proposed a TSK-17 cut-off of greater than 37 to identify high fear of movement.

Vlaeyen and Linton (2000) reviewed the evidence for fear-avoidance and its consequences in chronic musculoskeletal pain and formalised the cognitive-behavioural fear-avoidance model, in which catastrophic interpretation of pain leads to pain-related fear, avoidance and hypervigilance, and ultimately to disuse, depression and disability.

Leeuw et al. (2007) updated the review of the fear-avoidance model, confirming that pain-related fear is associated with greater disability and lower physical performance and that it can be targeted with exposure-based and educational interventions.

Gulrandhe et al. (2021) comprehensively discussed the fear-avoidance model of kinesiophobia, describing kinesiophobia as an irrational and incapacitating fear of movement arising from vulnerability to painful injury or re-injury. They noted that pain catastrophising and fear of movement heighten somatosensory awareness and the anticipation of worsening pain, and that kinesiophobia is associated with higher-order cognitive processes and anxiety. The TSK was presented as the principal assessment tool, with scores above 40 indicating high kinesiophobia. Conditions including ACL reconstruction, hip arthroplasty, hallux valgus and osteoporosis were discussed, in all of

which kinesiophobia complicated treatment and predicted poorer functional outcomes. Management recommendations included multimodal rehabilitation, psychological interventions (education, cognitive-behavioural therapy, perceptual stimulation and group work) and physical therapy, with the authors emphasising a multidirectional approach integrating psychological and physical treatment.

2.2 Stair Negotiation and its Biomechanics after TKR

Zeni and Snyder-Mackler (2010b) examined pre-operative predictors of persistent difficulty in stair ascent and descent after TKA, with particular emphasis on handrail use, in 105 patients. Pre-operative handrail use, age and quadriceps strength predicted handrail use at 3 months and 2 years after surgery, and many patients continued to use a handrail despite surgery. Younger patients with stronger quadriceps who did not use a handrail before surgery had the best outcomes. The authors stressed the importance of pre-operative functional status for predicting post-operative stair performance and for setting realistic patient expectations.

Smith et al. (2016) investigated stance time variability (STV) during stair ascent and descent before and after TKA in a pilot study of 13 patients with knee OA and age- and sex-matched healthy controls, using an instrumented staircase and motion analysis. STV during stair descent was significantly greater in the TKA group at 6 months after surgery, although the difference had disappeared by 12 months; no differences were found for stair ascent. Quadriceps strength remained lower in the TKA group, and although correlations between strength and STV were not significant, quadriceps weakness was considered a plausible contributor to increased variability. Patient-reported outcomes (LEFS, SF-36) improved after surgery, yet stair-climbing time remained longer than in controls even at 12 months, highlighting stair descent as a persistent challenge after TKA.

Rasnick et al. (2016) used musculoskeletal modelling to estimate knee joint loads and surrounding muscle forces during stair ascent in five patients with TKR and five healthy controls. Peak joint loads did not differ significantly between groups, but quadriceps forces, particularly those of rectus femoris and vastus lateralis, were lower in the TKR group, while knee flexor forces were higher. This compensatory muscle pattern produced joint loads similar to those of controls, indicating that muscle recruitment remains altered after TKR even when joint loading appears normal.

Lee et al. (2020) conducted a prospective cohort study of 84 patients to identify pre-operative physical factors predicting stair-climbing ability one month after TKA. Physical performance (stair-climbing test, 6MWT, TUG and isometric strength) was assessed before and after surgery. Post-operative stair-ascent ability correlated with

pre-operative TUG performance and knee extensor torque, while stair-descent ability was associated with pre-operative stair-descent performance and age. The authors concluded that balance and quadriceps strength are important determinants of stair-climbing capacity after TKA.

Kline et al. (2020) studied step-descent strategy after unilateral TKA in 40 participants (20 with TKA and 20 controls), assessing knee extensor, hip abductor and hip external rotator strength together with step-descent mechanics. Weakness of the knee extensors and hip external rotators of the operated limb persisted at 3 and 6 months after surgery. Patients altered their step-descent strategy bilaterally, using the hip musculature to unload the knee, and those with stronger hip muscles relied less on the knee support moment. The authors argued that current rehabilitation does not adequately address knee extensor and hip external rotator weakness, leading to compensatory strategies that impair stair performance.

Murakami et al. (2024) compared the difficulty of stair ascent and descent after TKR in 78 participants, analysing movement pattern and time taken. Stair descent was more difficult than ascent: more participants used a step-to pattern and handrail support and took longer to descend. The authors attributed this to the high demand of eccentric quadriceps contraction, co-contraction of knee extensors and flexors, and limited use of adjacent joints to compensate.

Asaki and Anan (2025) reported the case of an 80-year-old woman with persistent difficulty in stair negotiation after TKA despite rehabilitation. Assessment revealed restricted knee range of motion, quadriceps weakness and elevated fear of movement (TSK score 36). Surface EMG showed abnormal compensatory muscle activity suggestive of protective co-contraction driven by persistent fear. Out-patient physiotherapy improved range of motion and self-efficacy (increased PSEQ score), but fear and anxiety persisted and were associated with a prolonged swing phase during stair descent. The authors concluded that psychosocial factors such as fear, anxiety and depression adversely affect range-of-motion recovery and muscle recruitment after TKA and that rehabilitation should integrate psychological approaches with conventional physical therapy.

2.3 Prevalence and Determinants of Kinesiophobia after TKR

Desai et al. (2016) conducted a cross-sectional study of kinesiophobia in 120 knee arthroplasty patients (60 unilateral and 60 bilateral) using the TSK. Kinesiophobia was present in 56.67% of patients and was more common after unilateral than bilateral arthroplasty. High kinesiophobia was linked to fear of increased pain, of failing to achieve functional improvement, and to avoidance behaviour. Uncertainty, fear of not

reaching expected improvement and lack of confidence were important limiting factors. The authors noted that kinesiophobia is a modifiable factor that can be addressed through counselling and functional activity.

Mahajan et al. (2018) assessed the prevalence of kinesiophobia and its relationship with physiotherapy in 51 Indian patients after TKA using the TSK. Kinesiophobia was present in 64% of patients. Among those receiving physiotherapy, 45% had kinesiophobia, compared with 100% of those not receiving physiotherapy, a statistically significant difference ($p < 0.05$). Lack of physiotherapy, shorter rehabilitation, inadequate counselling and a history of trauma were identified as contributing factors. Patients with higher kinesiophobia had weaker quadriceps, lower range of motion and poorer movement patterns, resulting in poorer recovery and quality of life. Physiotherapy was concluded to act both physically and psychologically in reducing kinesiophobia.

Güney Yılmaz et al. (2023) investigated the influence of occupational self-perception on kinesiophobia after TKA in a prospective longitudinal study of 40 participants assessed pre-operatively and at 3 and 6 weeks post-operatively with the TSK, Occupational Self-Assessment (OSA), KOOS and TUG. Kinesiophobia was high both before and after surgery and peaked in the early post-operative period. TSK scores correlated negatively with OSA competence at all time points (pre-operative $r = -0.684$; 3 weeks $r = -0.717$; 6 weeks $r = -0.405$), whereas KOOS and TUG did not differ between high and low kinesiophobia groups. The authors concluded that a stronger sense of occupational competence protects against kinesiophobia and recommended that rehabilitation address occupational identity and self-efficacy in addition to pain and physical function.

Yan et al. (2023) used a prospective cohort design and latent class growth analysis in 257 patients to identify trajectories of kinesiophobia after TKA. Four trajectories were identified: low-stable, rapid-recovering, slow-recovering and moderate-stable. BMI, employment status, heart disease and pain severity predicted trajectory membership. At three months, rehabilitation outcomes differed significantly between trajectories, with the slow-recovering group showing greater psychological distress, poorer knee function and lower social participation, although activities of daily living did not differ. Kinesiophobia was therefore shown to be dynamic, and interventions should be tailored to the patient's trajectory.

Xing et al. (2025) examined factors associated with post-operative kinesiophobia in 446 elderly patients after TKA from a health-ecology perspective. The prevalence of kinesiophobia was 21.75%. Multivariate regression identified self-rehabilitation

efficacy, pain intensity, attention from children and social support as independent factors, and patients aged 80 years or more had significantly higher fear scores. Kinesiophobia was thus influenced by personal, behavioural, interpersonal, environmental and policy-level factors, and the authors recommended multifactorial interventions addressing psychological, social and environmental domains.

Du et al. (2025) performed a systematic review and meta-analysis of the prevalence and influencing factors of kinesiophobia after TKA, pooling 11 studies with 4,039 patients. They reported an overall prevalence of [insert pooled prevalence (%) and 95% CI from Du et al., 2025] and identified [insert principal influencing factors reported] as factors associated with kinesiophobia, underlining the need for routine screening after TKA.

Ren et al. (2026) used a longitudinal qualitative design, with semi-structured interviews of eight patients over three months, to explore how kinesiophobia develops after TKA. Three stages were identified: cognitive rigidity and avoidance; cognitive triggering and thawing; and cognitive reconstruction and attempt. Initially patients showed negative cognitions, self-objectification, dependence on external help and avoidance. With rehabilitation and growing self-efficacy they began to challenge catastrophic thoughts and change their behaviour. Central sensitisation of pain and poor health literacy were barriers, and improvement in self-efficacy was the key driver of change. Interventions should therefore be matched to the stage of the process and target self-efficacy and maladaptive cognitions.

2.4 Kinesiophobia and Functional Outcomes after TKR

Güney-Deniz et al. (2017) examined the influence of kinesiophobia on early functional outcomes in 46 patients at discharge after TKA, using the TSK, the 2-minute walk test (2-MWT), TUG, pain rating and knee flexion range of motion. Participants were grouped by level of kinesiophobia. Although TUG performance was similar, the low-kinesiophobia group walked further, reported less pain and had greater range of motion. TSK scores correlated strongly with pain ($r = 0.80$) and moderately with range of motion ($r = -0.47$) and walking distance ($r = -0.40$). Regression analysis showed that kinesiophobia explained 41% of the variance in walking distance, 47% in range of motion and 60% in pain. The authors identified fear of movement as a crucial psychological barrier in early recovery after TKA and recommended that clinicians address avoidance behaviour.

Brown et al. (2020) systematically reviewed 13 studies on the effect of kinesiophobia on outcomes after TKR. High TSK scores were consistently associated with poorer functional outcomes (WOMAC, Oxford Knee Score, 2- and 6-minute walk tests) for up to one year after surgery, whereas findings for pain were inconsistent. The relationship

between kinesiophobia and range of motion was consistently negative. Home-based functional exercise, cognitive-behavioural therapy and video-based psychological interventions were effective in reducing kinesiophobia. The authors concluded that both physical and psychological treatment are needed to overcome fear of movement.

2.5 Outcome Measures

2.5.1 Tampa Scale for Kinesiophobia (TSK-17)

The TSK is the most widely used instrument for measuring fear of movement/(re)injury (Kori et al., 1990; Vlaeyen et al., 1995). The original 17-item version is scored on a 4-point Likert scale from 1 (strongly disagree) to 4 (strongly agree); items 4, 8, 12 and 16 are reverse-scored, giving a total score of 17 to 68, with higher scores indicating greater kinesiophobia. Scores above 37 are commonly used to denote high kinesiophobia (Vlaeyen et al., 1995). The scale has demonstrated acceptable internal consistency (Cronbach's $\alpha = 0.68–0.91$), good test-retest reliability (ICC = 0.77–0.99) and construct and convergent validity in musculoskeletal populations, and it has been used extensively after TKR (Güney-Deniz et al., 2017; Brown et al., 2020).

2.5.2 Stair Climb Test (SCT)

The timed Stair Climb Test is one of the performance-based tests recommended by the Osteoarthritis Research Society International (OARSI) for people with hip or knee OA and after joint replacement (Dobson et al., 2013). It measures the time taken to ascend and descend a flight of stairs and reflects lower-limb strength, power, balance and stair-negotiation ability. The test has high test-retest reliability (ICC = 0.90–0.99), demonstrates construct validity through its associations with muscle strength, mobility and functional performance, and is responsive to change following rehabilitation (Dobson et al., 2013).

2.6 Summary of the Literature

The reviewed literature establishes that (i) stair negotiation, particularly descent, remains impaired for months after TKR and is characterised by step-to patterns, handrail dependence and altered muscle recruitment; (ii) physical factors such as quadriceps and hip strength explain only part of this impairment; (iii) kinesiophobia is common after TKR, more so after unilateral surgery, and is dynamic and multifactorial; and (iv) kinesiophobia is associated with poorer pain, range of motion and walking outcomes. However, few studies have directly correlated a validated measure of kinesiophobia with an objective measure of stair-climbing ability in patients with unilateral TKR, and evidence from India is scarce. The present study was designed to address this gap.

CHAPTER 3: METHODOLOGY

3.1 Study Design

An observational, cross-sectional, correlational study design was used.

3.2 Study Setting

The study was conducted at [name(s) of hospital / orthopaedic and physiotherapy department / rehabilitation centre], Jaipur, Rajasthan, in collaboration with the Department of Physiotherapy, Suresh Gyan Vihar University, Jaipur.

3.3 Study Duration

Data were collected over a period of [__ months], from [month, year] to [month, year].

3.4 Ethical Considerations

Ethical approval was obtained from the Institutional Ethics Committee of [name of committee] (Reference No. [____], dated [__/__/____]). The study was conducted in accordance with the Declaration of Helsinki and the ICMR National Ethical Guidelines for Biomedical and Health Research Involving Human Participants (2017). All participants received a participant information sheet in a language they understood (English or Hindi) and provided written informed consent before enrolment (Appendices B and C). Participation was voluntary, participants were free to withdraw at any time without any effect on their treatment, and confidentiality was maintained by coding all data.

3.5 Study Population

Target population: Individuals who have undergone Total Knee Replacement.

Accessible population: Individuals who have undergone primary unilateral TKR and are attending [study setting] for post-operative follow-up or physiotherapy.

3.6 Sampling Method

Non-probability convenience sampling was used. Consecutive eligible patients attending the study setting during the data-collection period were screened and invited to participate.

3.7 Sample Size

The sample size was calculated using G*Power software version 3.1.9.7 (Faul et al., 2009) for a bivariate normal correlation model (two-tailed), with an expected

correlation coefficient (effect size) of ρ = 0.40, α = 0.05 and power (1 − β) = 0.80. The minimum required sample size was 47 participants. Allowing for possible dropouts and incomplete data, a total of 50 participants were included

3.8 Selection Criteria

Inclusion criteria

1. Individuals who have undergone primary unilateral Total Knee Replacement.
2. Male and female patients aged 50–75 years.
3. Medically stable and able to ambulate independently, with or without an assistive device.
4. In the post-operative rehabilitation phase (4 weeks to 6 months after surgery).
5. Able to understand and follow verbal instructions.
6. Willing to participate and to provide written informed consent.

Exclusion criteria

1. Bilateral TKR (simultaneous or staged).
2. Revision TKR, or post-operative complications such as infection, wound dehiscence or implant failure.
3. Neurological disorders affecting balance or mobility (e.g., stroke, Parkinson's disease, peripheral neuropathy).
4. Severe cardiopulmonary conditions limiting physical activity.
5. Other lower-limb musculoskeletal conditions affecting stair climbing (e.g., hip fracture, contralateral severe knee pain, severe ankle pathology).
6. Cognitive impairment or psychiatric disorders interfering with assessment.

3.9 Variables

Type of variable	Variable	Measured by
Independent variable	Fear of movement (kinesiophobia)	Tampa Scale for Kinesiophobia (TSK-17)
Dependent variable	Stair-climbing ability	Stair Climb Test (SCT), time in seconds
Demographic / descriptive variables	Age, gender, height, weight, BMI, time since surgery, affected side, assistive device, handrail use, stair-negotiation pattern	Assessment proforma (Appendix D)

3.10 Materials and Instruments

- Participant information sheet and informed consent form (English/Hindi)
- Assessment proforma
- Tampa Scale for Kinesiophobia (TSK-17) questionnaire
- A standard flight of [12] stairs, step height approximately [17–18] cm, with handrails on [both sides / one side]
- Digital stopwatch
- Measuring tape and stadiometer; weighing scale
- Chair for rest, pen and data sheets

3.11 Outcome Measures

3.11.1 Tampa Scale for Kinesiophobia (TSK-17)

The TSK-17 is a 17-item self-report questionnaire measuring fear of movement/(re)injury (Kori et al., 1990; Miller et al., 1991). Each item is rated on a 4-point Likert scale (1 = strongly disagree, 2 = disagree, 3 = agree, 4 = strongly agree). Items 4, 8, 12 and 16 are reverse-scored (1 ↔ 4, 2 ↔ 3). The total score ranges from 17 to 68, with higher scores indicating greater fear of movement. A score above 37 was used to classify participants as having high kinesiophobia (Vlaeyen et al., 1995). The TSK has demonstrated internal consistency (Cronbach's α = 0.68–0.91), test-retest reliability (ICC = 0.77–0.99) and construct and convergent validity. The questionnaire (Appendix E) was administered face-to-face; for participants unable to read English, the items were read aloud in Hindi by the investigator, who recorded the responses without prompting.

Table 3.1: Interpretation of TSK-17 scores

TSK-17 total score	Interpretation
17–37	Low kinesiophobia
38–68	High kinesiophobia

3.11.2 Stair Climb Test (SCT)

The timed SCT, recommended by OARSI as a core performance-based test for people with knee OA and after joint replacement (Dobson et al., 2013), was used to measure stair-climbing ability. The participant ascended and descended a flight of [12] steps as quickly as safely possible, and the total time in seconds was recorded. Use of the handrail and of the participant's usual walking aid was permitted and recorded. Higher times indicate poorer stair-climbing ability. The SCT has high test-retest reliability

(ICC = 0.90–0.99), construct validity and responsiveness to rehabilitation. The detailed protocol is given in Appendix F.

In addition to time, the following observations were recorded during the SCT: stair-negotiation pattern during ascent and descent (reciprocal or step-to), leading limb, and handrail use (none, one hand, two hands).

3.12 Procedure

1. Permission was obtained from the authorities of the study setting and ethical clearance from the Institutional Ethics Committee.
2. Patients who had undergone unilateral TKR were screened against the selection criteria by reviewing their medical records and through a brief interview and examination.
3. The purpose and procedure of the study were explained to eligible patients, and written informed consent was obtained.
4. Demographic and clinical data (age, gender, height, weight, BMI, time since surgery, affected side and assistive device) were recorded on the assessment proforma.
5. Each participant completed the TSK-17 in a quiet room before any physical testing, so that the test performance would not influence the questionnaire responses.
6. After a seated rest of at least 5 minutes, the SCT was explained and demonstrated. One familiarisation trial was permitted, followed by a rest period, after which the timed trial was performed according to the protocol in Appendix F. The investigator guarded the participant throughout for safety.
7. The SCT time, stair-negotiation pattern and handrail use were recorded. Testing was stopped immediately if the participant reported severe pain, dizziness, chest discomfort or breathlessness, or appeared unsafe.
8. All data were coded and entered into the master chart (Appendix G) for statistical analysis.

Figure 3.1: Flow chart of the study procedure

Study flow
Ethical approval and permission from study setting
↓ Screening of patients after unilateral TKR (4 weeks – 6 months post-operative)

Study flow
↓ Application of inclusion and exclusion criteria
↓ Explanation of study and written informed consent
↓ Recording of demographic and clinical data
↓ Administration of TSK-17 (fear of movement)
↓ Rest (≥ 5 minutes) → familiarisation → timed Stair Climb Test
↓ Data entry, coding and statistical analysis
↓ Results, discussion and conclusion

3.13 Statistical Analysis

Data were entered into Microsoft Excel and analysed using [IBM SPSS Statistics version __ / other software].

Descriptive statistics

- Continuous variables (age, BMI, time since surgery, TSK-17 score and SCT time) were summarised as mean and standard deviation (and median with interquartile range where data were not normally distributed).
- Categorical variables (gender, affected side, kinesiophobia category, stair pattern and handrail use) were summarised as frequency and percentage.

Inferential statistics

- Normality of the continuous variables was assessed using the Shapiro–Wilk test.
- The correlation between TSK-17 score and SCT time was analysed using Pearson's product-moment correlation coefficient (r) if both variables were normally distributed, or Spearman's rank correlation coefficient (ρ) otherwise.
- As a supplementary analysis, SCT time was compared between participants with low (≤ 37) and high (> 37) kinesiophobia using the independent-samples t-test or the Mann–Whitney U test, depending on normality.
- The strength of correlation was interpreted according to Mukaka (2012) (Table 3.2).
- The level of statistical significance was set at $p < 0.05$. Results are presented in tables, graphs and charts.

Table 3.2: Interpretation of the correlation coefficient (Mukaka, 2012)

Size of correlation (|r|)	Interpretation
0.90 – 1.00	Very high correlation
0.70 – 0.90	High correlation
0.50 – 0.70	Moderate correlation
0.30 – 0.50	Low correlation
0.00 – 0.30	Negligible correlation

CHAPTER 4: RESULTS

> *TEMPLATE: The study data are not yet available. Every highlighted [ ] field in this chapter must be filled with the actual values obtained from the analysis. Do not report any value that was not obtained from the collected data. Delete this note before submission.*

A total of [__] patients were screened, of whom [__] met the selection criteria and [50] were included in the final analysis. [State reasons for exclusion/dropout, if any.] The results are presented below.

4.1 Demographic Characteristics

Table 4.1: Demographic characteristics of participants (continuous variables) (n = [50])

Variable	Mean ± SD	Minimum	Maximum
Age (years)	[ ]	[ ]	[ ]
Height (cm)	[ ]	[ ]	[ ]
Weight (kg)	[ ]	[ ]	[ ]
BMI (kg/m²)	[ ]	[ ]	[ ]
Time since surgery (weeks)	[ ]	[ ]	[ ]

Table 4.1 shows that the mean age of the participants was [__ ± __] years and the mean BMI was [__ ± __] kg/m². Participants were assessed at a mean of [__ ± __] weeks after surgery.

Table 4.2: Demographic characteristics of participants (categorical variables) (n = [50])

Variable	Category	Frequency (n)	Percentage (%)
Gender	Male	[ ]	[ ]
	Female	[ ]	[ ]
Affected side	Right	[ ]	[ ]
	Left	[ ]	[ ]
Assistive device	None	[ ]	[ ]
	Walking stick / walker	[ ]	[ ]

Of the [50] participants, [__ (__%)] were female and [__ (__%)] were male; the right knee was operated in [__ (__%)] participants (Table 4.2; Figures 4.1 and 4.2).

Figure 4.1: Gender distribution of participants

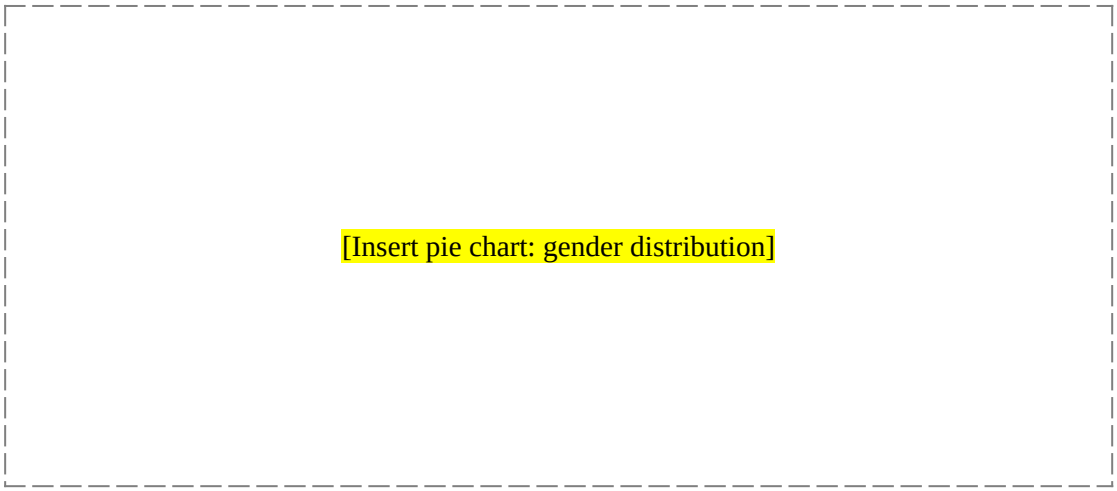
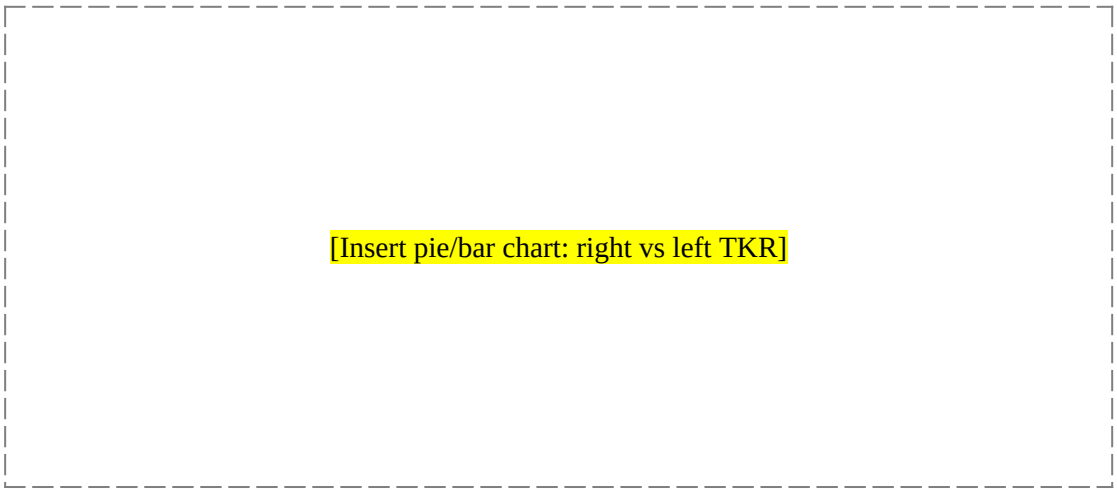


Figure 4.2: Distribution of affected side



4.2 Kinesiophobia and Stair-Climbing Ability

Table 4.3: Descriptive statistics of TSK-17 and SCT scores (n = [50])

Outcome measure	Mean ± SD	Median (IQR)	Range
TSK-17 score (17–68)	[ ]	[ ]	[ ]
SCT time (seconds)	[ ]	[ ]	[ ]

The mean TSK-17 score was [__ ± __], and the mean time to complete the SCT was [__ ± __] seconds (Table 4.3).

Table 4.4: Distribution of participants by level of kinesiophobia

Level of kinesiophobia	TSK-17 score	Frequency (n)	Percentage (%)
Low	17–37	[ ]	[ ]
High	38–68	[ ]	[ ]
Total		[50]	100

[__ (__%)] participants had high kinesiophobia and [__ (__%)] had low kinesiophobia (Table 4.4; Figure 4.3).

Figure 4.3: Distribution of participants by level of kinesiophobia

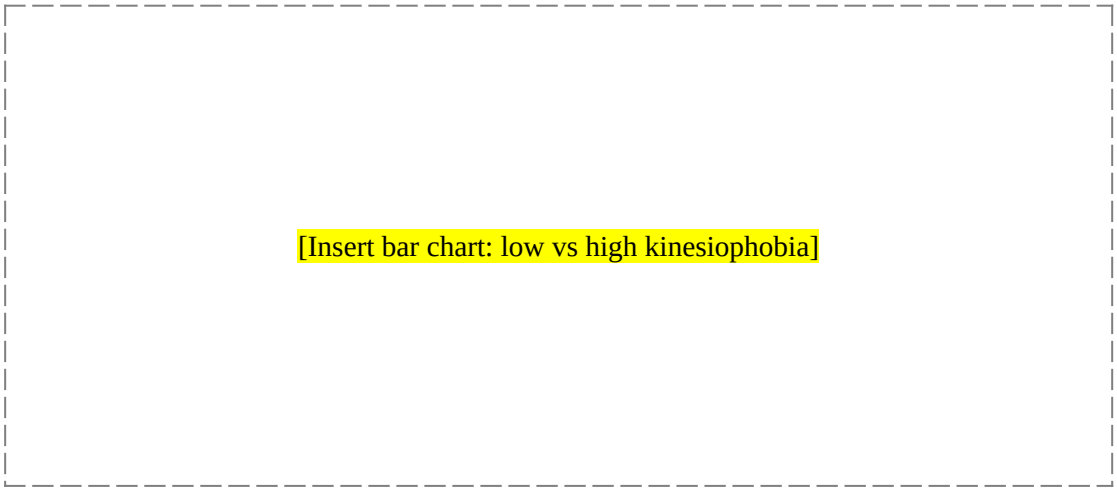


Table 4.5: Stair-negotiation strategy and handrail use during the SCT

Observation	Category	Ascent n (%)	Descent n (%)
Stair pattern	Reciprocal	[ ]	[ ]
	Step-to	[ ]	[ ]
Handrail use	None	[ ]	[ ]
	One hand	[ ]	[ ]
	Two hands	[ ]	[ ]

[Describe Table 4.5, e.g., “A step-to pattern was used by __% of participants during descent compared with __% during ascent.”]

4.3 Test of Normality

Table 4.6: Shapiro–Wilk test of normality

Variable	W statistic	df	p-value	Distribution
TSK-17 score	[ ]	[ ]	[ ]	[Normal / Non-normal]
SCT time	[ ]	[ ]	[ ]	[Normal / Non-normal]

[e.g., “Both variables were normally distributed ($p > 0.05$); therefore Pearson’s correlation coefficient was used.” OR “SCT time was not normally distributed ($p < 0.05$); therefore Spearman’s rank correlation coefficient was used.”]

4.4 Correlation between Kinesiophobia and Stair-Climbing Ability

Table 4.7: Correlation between kinesiophobia (TSK-17) and stair-climbing ability (SCT)

Variables	Correlation coefficient (r / ρ)	p-value	Interpretation
TSK-17 score vs SCT time	[ ]	[ ]	[e.g., Moderate positive, significant]

Table 4.7 shows a [negligible / low / moderate / high] [positive / negative] correlation between TSK-17 score and SCT time (r = [__], p = [__]). [Because $p < 0.05$, the null hypothesis is rejected and the alternative hypothesis is accepted. / Because $p \geq 0.05$, the null hypothesis is retained.] A positive coefficient indicates that participants with greater fear of movement took longer to complete the SCT, i.e., had poorer stair-climbing ability (Figure 4.4).

Figure 4.4: Scatter plot of TSK-17 score versus SCT time

[Insert scatter plot: TSK-17 (x-axis) vs SCT time in seconds (y-axis) with regression line and r value]

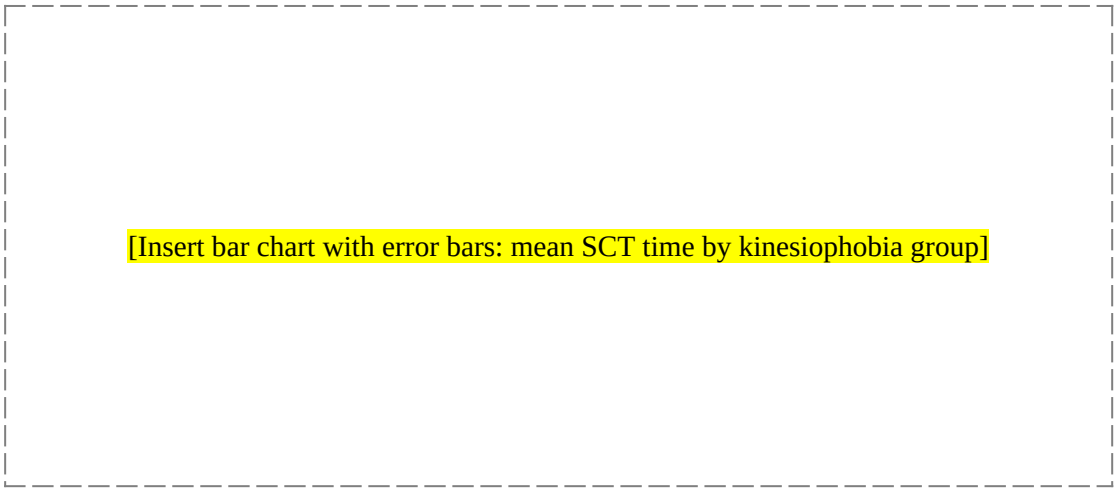
4.5 Comparison of Stair-Climbing Ability by Level of Kinesiophobia

Table 4.8: Comparison of SCT time between low and high kinesiophobia groups

Group	n	SCT time (s), Mean ± SD	Test statistic (t / U)	p-value
Low kinesiophobia (≤ 37)	[ ]	[ ]	[ ]	[ ]
High kinesiophobia (> 37)	[ ]	[ ]		

[Describe Table 4.8, e.g., “Participants with high kinesiophobia took significantly longer to complete the SCT than those with low kinesiophobia (__ ± __ s vs __ ± __ s; p = __).”]

Figure 4.5: Mean SCT time in low and high kinesiophobia groups



CHAPTER 5: DISCUSSION

> *DRAFT: This discussion has been written for the anticipated finding of a significant positive correlation between TSK-17 score and SCT time. Once the results are available, replace all highlighted fields and revise the wording wherever the actual findings differ (e.g., if the correlation is weak or non-significant). Delete this note before submission.*

The present cross-sectional study was undertaken to determine the correlation between fear of movement (kinesiophobia) and stair-climbing ability in individuals following unilateral TKR. Kinesiophobia was measured with the TSK-17 and stair-climbing ability with the timed SCT in [50] participants assessed between 4 weeks and 6 months after surgery.

5.1 Level of Kinesiophobia after Unilateral TKR

The mean TSK-17 score in the present study was [__ ± __], and [__%] of participants had high kinesiophobia (TSK-17 > 37). This proportion is [comparable to / higher than / lower than] the prevalence of 56.67% reported by Desai et al. (2016) in Indian knee arthroplasty patients and the 64% reported by Mahajan et al. (2018), and [higher / lower] than the 21.75% reported by Xing et al. (2025) in elderly Chinese patients. Differences in prevalence across studies may be explained by differences in the time since surgery, the TSK version and cut-off used (e.g., > 37 versus > 40, Gulrandhe et al., 2021), age, access to physiotherapy and cultural beliefs about pain and movement.

The relatively [high] level of kinesiophobia in this unilateral TKR sample is consistent with the observation that unilateral patients report more fear than bilateral patients (Desai et al., 2016; Kazarian et al., 2021). Having a pain-free non-operated limb allows patients to offload the operated knee, so that fearful beliefs are never disconfirmed by successful loading. In addition, the study population was assessed in the early post-operative period, when kinesiophobia has been shown to peak (Güney Yılmaz et al., 2023) and before many patients have moved into a “rapid-recovering” trajectory (Yan et al., 2023).

5.2 Stair-Climbing Ability after Unilateral TKR

The mean SCT time was [__ ± __] seconds, and [__%] of participants used a step-to pattern during descent compared with [__%] during ascent. These findings agree with Murakami et al. (2024), who found stair descent to be more difficult than ascent after TKR, with greater use of step-to patterns, handrails and longer descent times. The greater difficulty of descent reflects the high demand for eccentric quadriceps control and the co-contraction strategies adopted to stabilise the knee (Smith et al., 2016; Kline et al., 2020). The frequent use of handrails in the present sample [__%] is also in line

with Zeni and Snyder-Mackler (2010b), who reported persistent handrail use in many patients up to two years after surgery.

5.3 Correlation between Kinesiophobia and Stair-Climbing Ability

The principal finding of this study is a [moderate], [statistically significant] positive correlation between TSK-17 score and SCT time (r = [__], p = [__]): participants with greater fear of movement took longer to climb and descend stairs. The null hypothesis was therefore [rejected]. In addition, participants with high kinesiophobia took [significantly] longer to complete the SCT than those with low kinesiophobia.

This finding is consistent with the fear-avoidance model (Vlaeyen & Linton, 2000; Leeuw et al., 2007), which predicts that catastrophic interpretation of pain leads to fear, avoidance and hypervigilance and, through disuse, to reduced physical performance. It extends the findings of Güney-Deniz et al. (2017), who showed that kinesiophobia explained a substantial proportion of the variance in walking distance and range of motion after TKA, and of Brown et al. (2020), whose systematic review found consistently poorer functional outcomes in patients with high TSK scores. The present study adds that the association also applies to a specific, high-demand task, namely stair negotiation, measured with an objective performance test rather than a questionnaire.

Several mechanisms may explain the association. First, fear of movement may alter motor control directly. Asaki and Anan (2025) documented protective co-contraction and a prolonged swing phase during stair descent in a patient with persistent fear despite improved range of motion, indicating that fear can impair stair performance independently of structural factors. Bracing the knee through co-contraction of the quadriceps, hamstrings and gastrocnemius reduces the knee flexion needed for a smooth descent, slows the movement and encourages a step-to pattern. Second, fearful patients may avoid stairs and limit their participation in exercise, leading to persistent quadriceps and hip weakness (Kline et al., 2020; Mahajan et al., 2018), which in turn prolongs stair-climbing time. Third, in unilateral TKR the availability of a healthy limb allows consistent offloading of the operated knee (Song et al., 2024), so that the patient never relearns eccentric control of the replaced knee during descent.

Physical factors such as quadriceps strength, balance and pre-operative function are known predictors of post-operative stair ability (Zeni & Snyder-Mackler, 2010b; Lee et al., 2020). The present findings suggest that psychological factors should be considered alongside these physical determinants, which is consistent with the observation that physical impairments alone explain only part of the variability in stair performance after TKR (Vinueza-Veloz et al., 2014).

5.4 Clinical Implications

- Kinesiophobia should be screened routinely after TKR using a brief, validated tool such as the TSK, particularly in patients who are slow to progress with stair training.
- Patients with high TSK scores may benefit from pain neuroscience education, reassurance about the safety of loading the prosthetic knee, and goal-setting that targets self-efficacy (Ren et al., 2026; Xing et al., 2025).
- Graded exposure to stair negotiation (progressing step height, number of steps, reduced handrail use and reciprocal patterns) may help to disconfirm fearful beliefs while also building strength and motor control.
- Rehabilitation should combine conventional strengthening of the quadriceps and hip musculature with cognitive-behavioural strategies, in line with a biopsychosocial approach (Gulrandhe et al., 2021; Brown et al., 2020).
- Involving family members and providing adequate counselling and access to physiotherapy may reduce kinesiophobia, particularly in the Indian context (Mahajan et al., 2018; Xing et al., 2025).

5.5 Limitations of the Study

1. The cross-sectional design does not allow causal inferences; it cannot determine whether fear leads to poor stair performance or poor performance leads to fear.
2. Convenience sampling from [a single centre] in Jaipur limits the generalisability of the findings.
3. The relatively wide post-operative window (4 weeks to 6 months) may have introduced variability, as both kinesiophobia and stair ability change over time.
4. Potential confounders such as pain intensity, quadriceps strength, knee range of motion, pre-operative function, anxiety, depression and pain catastrophising were not measured or controlled.
5. Only the total SCT time was recorded; ascent and descent were not timed separately, and kinematic or electromyographic data were not collected.
6. The TSK-17 was administered in English or read aloud in Hindi; [a formally validated Hindi version was / was not] used.

5.6 Recommendations for Future Research

- Longitudinal studies following patients from before surgery to one year after TKR to examine how changes in kinesiophobia relate to changes in stair performance.
- Studies with larger, multi-centre samples, and comparison of unilateral and bilateral TKR.
- Multivariate analyses including pain, quadriceps strength, range of motion, self-efficacy and catastrophising to determine the independent contribution of kinesiophobia.
- Separate timing of stair ascent and descent and the use of motion analysis or surface EMG to examine fear-related changes in movement.
- Randomised controlled trials of interventions targeting kinesiophobia (education, graded exposure, cognitive-behavioural therapy) on stair-climbing outcomes.

CHAPTER 6: CONCLUSION

DRAFT: Revise according to the actual findings and delete this note.

The present study concludes that there is a [moderate, statistically significant positive] correlation between fear of movement (kinesiophobia), measured with the TSK-17, and stair-climbing time, measured with the Stair Climb Test, in individuals 4 weeks to 6 months after unilateral Total Knee Replacement. Individuals with higher kinesiophobia demonstrated poorer stair-climbing ability. The alternative hypothesis is therefore [accepted].

Kinesiophobia was present in [__%] of participants, indicating that fear of movement is common after unilateral TKR. These findings support a biopsychosocial approach to post-operative rehabilitation in which kinesiophobia is routinely screened for and addressed alongside physical impairments, in order to improve stair negotiation, functional independence and patient satisfaction after TKR.

SUMMARY

Total Knee Replacement is an effective treatment for end-stage knee osteoarthritis, but many patients continue to experience difficulty with demanding functional tasks such as stair negotiation. According to the fear-avoidance model, fear of movement (kinesiophobia) may lead to avoidance, disuse and disability independently of physical impairment. Patients with unilateral TKR may be particularly susceptible, because the non-operated limb allows them to avoid loading the replaced knee.

This cross-sectional study aimed to determine the correlation between kinesiophobia and stair-climbing ability following unilateral TKR. [50] participants aged 50–75 years who were 4 weeks to 6 months after unilateral TKR were recruited by convenience sampling from [study setting]. Kinesiophobia was assessed with the Tampa Scale for Kinesiophobia (TSK-17) and stair-climbing ability with the timed Stair Climb Test. Data were analysed using [Pearson's / Spearman's] correlation coefficient with significance set at $p < 0.05$.

The mean TSK-17 score was [__ ± __] and [__%] of participants had high kinesiophobia. The mean SCT time was [__ ± __] seconds. A [strength and direction] correlation was found between TSK-17 score and SCT time (r = [__], p = [__]). [One-sentence conclusion.]

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APPENDICES

Appendix A: Ethical Clearance Certificate

Insert a scanned copy of the Institutional Ethics Committee approval letter here.

Appendix B: Participant Information Sheet

Study title: Correlation between fear of movement (kinesiophobia) and stair-climbing ability following unilateral Total Knee Replacement: a cross-sectional study

Investigator: Kshma Sahu, MPT student, Department of Physiotherapy, Suresh Gyan Vihar University, Jaipur. Contact: [phone number]

Guide: Dr. Sheenam Popli, Associate Professor, Department of Physiotherapy, Suresh Gyan Vihar University, Jaipur.

Purpose of the study: You are invited to take part in a research study that aims to find out whether fear of movement is related to the ability to climb stairs after knee replacement surgery on one knee.

What you will be asked to do: If you agree to participate, you will be asked some questions about yourself and your surgery, and to complete a short questionnaire of 17 statements about your feelings regarding movement and pain (about 10 minutes). You will then be asked to walk up and down a flight of stairs at a fast but safe pace while being timed. You may use the handrail and your usual walking aid. The total time required is about 30 minutes.

Risks and discomforts: Stair climbing may cause mild temporary knee discomfort or fatigue. A physiotherapist will stay beside you throughout to ensure your safety, and you may rest or stop at any time.

Benefits: There is no direct benefit to you. The results may help physiotherapists to improve rehabilitation for future patients after knee replacement.

Confidentiality: Your name will not appear in any report. All information will be coded and kept confidential and used only for research purposes.

Voluntary participation: Your participation is entirely voluntary. You may refuse or withdraw at any time without giving a reason, and this will not affect your treatment in any way. There is no cost to you and no payment for participation.

Appendix C: Informed Consent Form

Participant ID: _____

1. I confirm that I have read (or have had read to me) and understood the participant information sheet for the above study and have had the opportunity to ask questions.
2. I understand that my participation is voluntary and that I am free to withdraw at any time without giving any reason and without my medical care being affected.
3. I understand that my personal information will be kept confidential and that the data may be used for research and publication without revealing my identity.
4. I agree to take part in the above study.

Name of participant: _____

Signature / thumb impression:

_____

Date: _____

Name of witness: _____

Signature of witness: _____

Name of investigator: _____

Signature of investigator: _____

सूचित सहमति पत्र (Hindi)

प्रतिभागी आईडी: _____

1. मैंने उपरोक्त अध्ययन की प्रतिभागी सूचना पत्र को पढ़ लिया है (या मुझे पढ़कर सुनाया गया है) और समझ लिया है, तथा मुझे प्रश्न पूछने का अवसर दिया गया है।
2. मैं समझता/समझती हूँ कि मेरी भागीदारी स्वैच्छिक है और मैं किसी भी समय, बिना कोई कारण बताए, अध्ययन से हट सकता/सकती हूँ, और इससे मेरे उपचार पर कोई प्रभाव नहीं पड़ेगा।
3. मैं समझता/समझती हूँ कि मेरी व्यक्तिगत जानकारी गोपनीय रखी जाएगी और मेरी पहचान उजागर किए बिना आँकड़ों का उपयोग शोध एवं प्रकाशन के लिए किया जा सकता है।
4. मैं इस अध्ययन में भाग लेने के लिए सहमत हूँ।

प्रतिभागी का नाम: _____

हस्ताक्षर / अँगूठे का निशान: _____

दिनांक: _____

गवाह का नाम एवं हस्ताक्षर: _____

शोधकर्ता का नाम एवं हस्ताक्षर:

_____

Appendix D: Assessment Proforma

Participant ID: _____ Date of assessment: ____/____/_____

Item	Details
Name (initials)	
Age (years)	
Gender	Male / Female
Height (cm)	
Weight (kg)	
BMI (kg/m²)	
Occupation	
Type of residence	Single storey / Multi-storey; stairs at home: Yes / No
Date of surgery	
Time since surgery (weeks)	
Operated side	Right / Left
Assistive device	None / Stick / Walker / Other
Currently receiving physiotherapy	Yes / No; sessions per week: ____
Relevant medical history	
Medications	
Eligibility confirmed	Inclusion criteria met: Yes / No; Exclusion criteria absent: Yes / No
TSK-17 total score	_____ / 68 (Low ≤ 37 / High > 37)
SCT time (seconds)	_____
Ascent pattern / leading limb	Reciprocal / Step-to; leading: Operated / Non-operated
Descent pattern / leading limb	Reciprocal / Step-to; leading: Operated / Non-operated
Handrail use	None / One hand / Two hands
Adverse events / remarks	

Appendix E: Tampa Scale for Kinesiophobia (TSK-17)

Instructions: This is a list of statements about how people feel about movement and pain. Please indicate how much you agree or disagree with each statement by ticking one box. Think about how you feel *now*, about your operated knee.

No.	Statement	Strongly disagree (1)	Disagree (2)	Agree (3)	Strongly agree (4)
1	I'm afraid that I might injure myself if I exercise.	☐	☐	☐	☐
2	If I were to try to overcome it, my pain would increase.	☐	☐	☐	☐
3	My body is telling me I have something dangerously wrong.	☐	☐	☐	☐
4 *	My pain would probably be relieved if I were to exercise.	☐	☐	☐	☐
5	People aren't taking my medical condition seriously enough.	☐	☐	☐	☐
6	My accident has put my body at risk for the rest of my life.	☐	☐	☐	☐
7	Pain always means I have injured my body.	☐	☐	☐	☐
8 *	Just because something aggravates my pain does not mean it is dangerous.	☐	☐	☐	☐
9	I am afraid that I might injure myself accidentally.	☐	☐	☐	☐
10	Simply being careful that I do not make any unnecessary movements is the safest thing I can do to prevent my pain from worsening.	☐	☐	☐	☐
11	I wouldn't have this much pain if there weren't something potentially dangerous going on in my body.	☐	☐	☐	☐
12 *	Although my condition is painful, I would be better off if I were physically active.	☐	☐	☐	☐
13	Pain lets me know when to stop exercising so that I don't injure myself.	☐	☐	☐	☐
14	It's really not safe for a person with a condition like mine to be physically active.	☐	☐	☐	☐

No.	Statement	Strongly disagree (1)	Disagree (2)	Agree (3)	Strongly agree (4)
15	I can’t do all the things normal people do because it’s too easy for me to get injured.	☐	☐	☐	☐
16*	Even though something is causing me a lot of pain, I don’t think it’s actually dangerous.	☐	☐	☐	☐
17	No one should have to exercise when he/she is in pain.	☐	☐	☐	☐

Reverse-scored items (4, 8, 12, 16): 1 = 4, 2 = 3, 3 = 2, 4 = 1. Total score = sum of all 17 items (range 17–68). Score > 37 = high kinesiophobia

Source: Miller, Kori & Todd (1991); Vlaeyen et al. (1995).

Appendix F: Stair Climb Test – Protocol and Recording Sheet

Set-up

- Staircase of [12] steps, each approximately [17–18] cm high, with handrail(s). Record the actual number and height of steps: _____ steps × _____ cm.
- Well-lit, uncluttered environment; participant wears usual footwear.
- A chair is placed at the bottom of the stairs; a digital stopwatch is used for timing.

Instructions to the participant

"When I say 'Go', walk up the stairs, turn around at the top, and come back down to the bottom as quickly as you can while staying safe. You may use the handrail and your walking aid if you need to."

Procedure

1. Check resting status and ask about pain and any symptoms; do not test if the participant feels unwell.
2. Demonstrate the test and allow one familiarisation trial, followed by a rest of at least 2 minutes.
3. Start the stopwatch on "Go" and stop it when both feet are back on the floor at the bottom of the stairs.
4. Guard the participant from behind and slightly to the side during ascent and in front during descent.
5. Record the total time to the nearest 0.01 second, the stepping pattern, the leading limb and handrail use.
6. Stop the test immediately in case of severe pain, dizziness, chest pain, breathlessness or unsafe performance.

Recording sheet

Parameter	Ascent	Descent
Pattern (reciprocal / step-to)		
Leading limb (operated / non-operated)		
Handrail (none / one / two hands)		
Walking aid used		

Total SCT time: _____ seconds **Remarks:**

_____

Appendix G: Master Chart

Enter one row per participant (n = 50). Wks = weeks post-op; Grp = TSK group; Asc/Desc = stair pattern; Rail = handrail use. Codes: Gender M = 1, F = 2; Side R = 1, L = 2; Pattern reciprocal = 1, step-to = 2; Handrail none = 0, one hand = 1, two hands = 2; Kinesiophobia low = 1, high = 2.

ID	Age	Sex	BMI	Wks	Side	TSK	Grp	SCT (s)	Asc	Desc	Rail
1											
2											
3											
4											
5											
6											
7											
8											
9											
10											
…											
50											

Appendix H: Plagiarism Report

> *Insert the plagiarism check certificate issued by the University library (e.g., Turnitin / Drillbit / Ouriginal) here.*